Early Forecasting of Catastrophic Forgetting in CLIP Fine-Tuning

Saanvi Eha Chandra*
Nexus Labs
Palo Alto, USA
Saanvi.eha.chandra@gmail.com

Avni Gautam*
Nexus Labs
Palo Alto, USA
gautamsumit@gmail.com

Krisha Gupta*
Nexus Labs
Palo Alto, USA
krishag1016@gmail.com

Neel Ranadive*
Nexus Labs
Palo Alto, USA
ranadiveneel@gmail.com

Aarav Shah*
Nexus Labs
Palo Alto, USA
arvjcs@gmail.com

Noah Yoo*
Nexus Labs
Palo Alto, USA
nynoahyoo@gmail.com

Srikanth Samy
University of California, Berkeley
Berkeley, USA
ssamy@berkeley.edu

Ishaan Gupta
Stanford University
Stanford, USA
ishaangupta@stanford.edu

Abstract—Fine-tuning a vision–language model on a narrow dataset improves the target task while eroding the zero-shot competence that made the model valuable. We ask whether the final magnitude of that erosion can be forecast from a short warm-up of the fine-tuning run. Our experiment comprises 48 fine-tuning runs of CLIP ViT-B/32 (16 specialised datasets × 3 learning rates × 20 epochs), logging five trajectory signals after every epoch and measuring forgetting as the mean drop in zero-shot accuracy over CIFAR-100, CIFAR-10, and STL-10. Forecasters are evaluated by leave-one-dataset-out cross-validation, so that all three learning-rate runs of a held-out dataset are unseen. Four cheap pre-training descriptors alone reach $R^2 = 0.43$; adding a five-epoch warm-up raises this to $R^2 = 0.92$ and cuts mean absolute error from 0.070 to 0.025. Simple tabular regressors beat both an MLP ($R^2 = -0.66$) and an LSTM ($R^2 = 0.81$) in this 48-row, 16-group regime. Critically, we add the persistence control that the underlying study identified as missing: predicting final forgetting to equal forgetting at the end of the warm-up already attains $R^2 = 0.85$ and Spearman $\rho = 0.945$. The learned forecaster's advantage over naive curve continuation is therefore real but narrow, and it is confined to very short warm-ups ($K \leq 5$); beyond $K \approx 6$ the two are indistinguishable. We conclude that most of the "forecasting" signal is trajectory persistence, and that future work in this area must report a persistence baseline to make a meaningful claim.

Keywords—Catastrophic forgetting, CLIP, fine-tuning dynamics, grouped cross-validation, multimodal learning, transfer learning, zero-shot classification.

I. Introduction

Contrastively pretrained vision–language models such as CLIP [1] derive their practical value from breadth: a single set of weights classifies arbitrary categories by comparing image embeddings against text prompts, with no task-specific training. Practitioners routinely trade that breadth for depth by fine-tuning on a specialised corpus. The weights that encode the specialised task are the same weights that encoded everything else, and updating them overwrites prior competence—the phenomenon known as catastrophic forgetting [2], [3].

The economics are unfavourable. A fine-tuning campaign is paid for in advance, but the damage is measured afterwards. If the eventual loss of general capability could be estimated from a short prefix of the run, a practitioner could abort, reduce the learning rate, or switch to a robustness-preserving procedure such as weight-space interpolation [4], [5] before most of the compute is spent.

Prior work predicts forgetting from static properties of the pretrained model and target dataset. We ask a different question:

> Given the first K epochs of a fine-tuning run on a dataset never seen before, how accurately can the final forgetting be forecast—and how much of that accuracy is anything more than extrapolating the curve?

The second clause is the part most easily neglected. A forecaster that consumes Φ_K, the forgetting already observed at epoch K, as an input feature is in competition not with an uninformed prior but with the trivial rule $\hat{\Phi}_T = \Phi_K$. We report that comparison explicitly, and it substantially tempers the headline.

A. Contributions

1) A cached, fully reproducible 48-run fine-tuning campaign that logs five trajectory signals per epoch across 16 datasets spanning three visual domains and three learning rates.
2) A grouped (leave-one-dataset-out) forecasting benchmark in which every normalisation and model fit occurs inside the fold, comparing five estimator families under a static-only versus static-plus-warm-up ablation.
3) A persistence control. We show that $\hat{\Phi}_T = \Phi_K$ reaches $R^2 = 0.85$, so that the learned forecaster's $R^2 = 0.92$ represents a 0.07 improvement over curve continuation rather than the 0.49 improvement suggested by the static-only ablation alone.
4) A characterisation of where learning helps: the learned model dominates persistence only for $K \leq 5$, the regime where aborting a run still saves meaningful compute.

II. Experimental Design

A. Model and forgetting definition

All runs start from CLIP ViT-B/32 with the original OpenAI pretrained weights (151.3M parameters). Zero-shot classification follows the standard prompt-ensembling recipe: for class names $c_1, \ldots, c_M$ we build prompts "a photo of a c_m.", encode them with the text tower, ℓ_2-normalise, and predict

$$\hat{y}(\mathbf{x}) \;=\; \arg\min_m \;-\frac{f_{\text{img}}(\mathbf{x})^\top g_{\text{txt}}(c_m)}{\|f_{\text{img}}(\mathbf{x})\|\;\|g_{\text{txt}}(c_m)\|}. \tag{1}$$

Let $\mathcal{B} = \{\text{CIFAR-100}, \text{CIFAR-10}, \text{STL-10}\}$ be held-out general benchmarks, and $A_b^{(0)}$ the pretrained zero-shot accuracy on $b \in \mathcal{B}$. We measure $A_b^{(0)} = 0.617$, 0.861, and 0.956 respectively. Forgetting after epoch t is the mean accuracy loss,

$$\Phi_t \;=\; \frac{1}{|\mathcal{B}|}\sum_{b\in\mathcal{B}}\left(A_b^{(0)} - A_b^{(t)}\right). \tag{2}$$

$\Phi_t > 0$ denotes lost general capability; $\Phi_t < 0$ would denote improvement. The prediction target is Φ_T with $T = 20$.

B. Fine-tuning protocol

For each dataset d and learning rate η we reset the encoder to pretrained weights, attach a freshly initialised linear head $W_d \in \mathbb{R}^{K_d \times 512}$, and minimise cross-entropy over both the image tower and the head with AdamW [6]:

$$\mathcal{L}_{\text{ft}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\big(W_{d,y_i}^\top f_{\text{img}}(\mathbf{x}_i)\big)}{\sum_k \exp\big(W_{d,k}^\top f_{\text{img}}(\mathbf{x}_i)\big)}. \tag{3}$$

The encoder uses learning rate $\eta \in \{10^{-6}, 5\times10^{-6}, 10^{-5}\}$, the head uses 10^{-3}, weight decay is 10^{-4}, batch size is 128, and every dataset is subsampled to an identical budget of 4000 training images so that a large corpus cannot appear to cause more forgetting merely by supplying more gradient steps. This yields $16 \times 3 = 48$ runs of 20 epochs.

C. Datasets

The 16 target datasets fall into three families by visual distance from web photography: natural (OxfordPets, Food101, FGVCAircraft, DTD, EuroSAT), digits (SVHN, MNIST, GTSRB), and medical (eight MedMNIST collections [7]). The medical and digit sets are natively 28×28 or 32×32 and must be upsampled roughly eightfold to CLIP's 224×224 input, which introduces a systematic blur absent from the natural family.

D. Logged signals

After every epoch we record five quantities. Beyond Φ_t and the training loss and accuracy, two measure how far the network has physically moved from its initialisation:

$$\delta_t^{\text{par}} = \frac{\|\boldsymbol{\theta}_t - \boldsymbol{\theta}_0\|_2}{\|\boldsymbol{\theta}_0\|_2}, \tag{4}$$

$$\delta_t^{\text{emb}} = \mathbb{E}_{\mathbf{x}\sim\mathcal{P}}\left[1 - \cos\big(f_t(\mathbf{x}),\, f_0(\mathbf{x})\big)\right], \tag{5}$$

where $\mathcal{P}$ is a fixed probe set of general images. Equation (4) is parameter drift, (5) is representation drift. Both are plausible early proxies for damage, because forgetting is ultimately a consequence of the encoder's outputs moving away from the geometry the text tower was aligned to.

III. Observed Forgetting

A. Trajectories

Figure 1 shows all 48 forgetting trajectories. Two effects are immediate and large. First, forgetting scales sharply with the learning rate: at $\eta = 10^{-6}$ almost every dataset ends below $\Phi_{20} = 0.10$, whereas at $\eta = 10^{-5}$ several exceed 0.35, i.e. more than a third of the pretrained general accuracy is destroyed. Second, curves are monotone and saturating for essentially every run, which is precisely the property that later makes naive persistence a strong forecaster.

B. Which datasets forget most

Figure 2 ranks final forgetting at the largest learning rate and summarises it by family. Family means are 0.309 (digits), 0.283 (medical), and 0.201 (natural), with ranges given in Table I. The ordering partially matches the naive expectation that visually distant data causes more damage, but the medical family spans almost the entire observed range (0.147 to 0.382), so "distance from web photos" is not by itself a sufficient predictor.

IV. Forecasting Setup

A. Features

Each of the 48 runs becomes one row. The static block $\mathbf{s}_d$ contains four descriptors computed before any training, plus $\log_{10}\eta$:

- semantic_dist: $1-$ pretrained zero-shot accuracy on d;
- clip_embed_dist: distance between the mean CLIP embedding of d and that of general photography;
- spectral_dist: a model-free Fourier-domain texture discrepancy;
- frechet_dist: a Fréchet distribution distance in embedding space [8].

The warm-up block $\mathbf{w}_d^{(K)}$ summarises the first K epochs by the terminal value of each logged signal together with the fitted slope of the forgetting curve,

$$\hat{\beta}_K = \frac{\sum_{t=1}^{K}(t-\bar{t})\left(\Phi_t - \bar{\Phi}\right)}{\sum_{t=1}^{K}(t-\bar{t})^2}. \tag{6}$$

Figure 3 (a) shows that individual static descriptors correlate only weakly with the outcome ($|r| \le 0.44$, $n = 16$), motivating multivariate models; panel (b) shows the co-evolution of the logged signals within a single run.

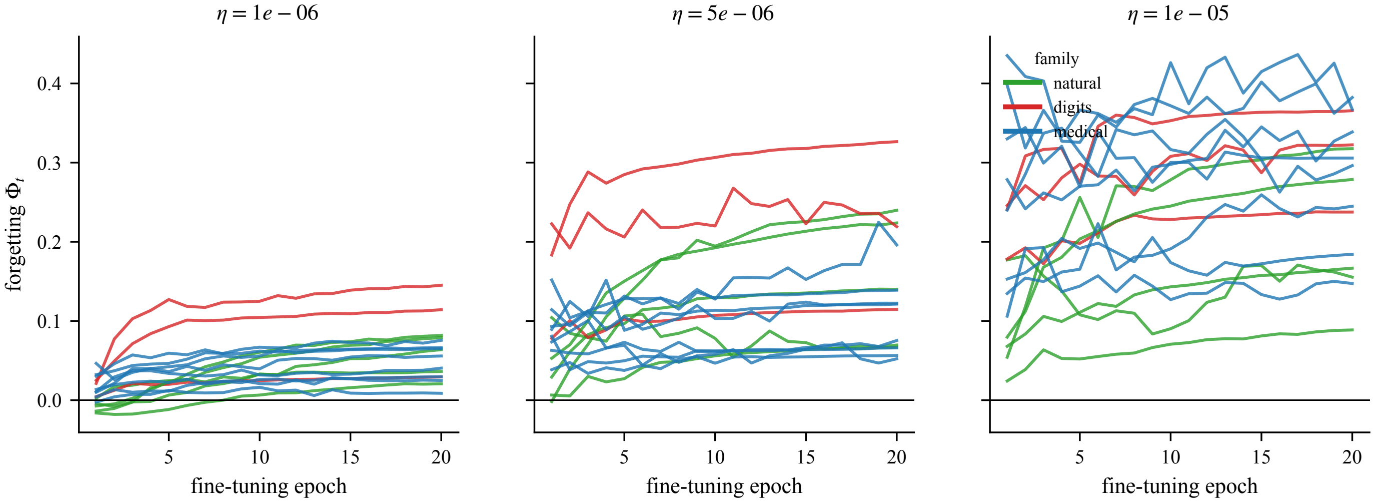

Fig. 1. Forgetting Φ_t during fine-tuning for all 16 datasets at the three learning rates. Colour encodes dataset family. Curves are near-monotone and saturating, which is the structural reason a persistence forecaster performs well.

TABLE I
Final forgetting Φ_{20} at $\eta = 10^{-5}$, grouped by dataset family.

Family	Datasets	Mean	Min	Max
digits	3	0.309	0.238	0.366
medical	8	0.283	0.147	0.382
natural	5	0.201	0.089	0.318

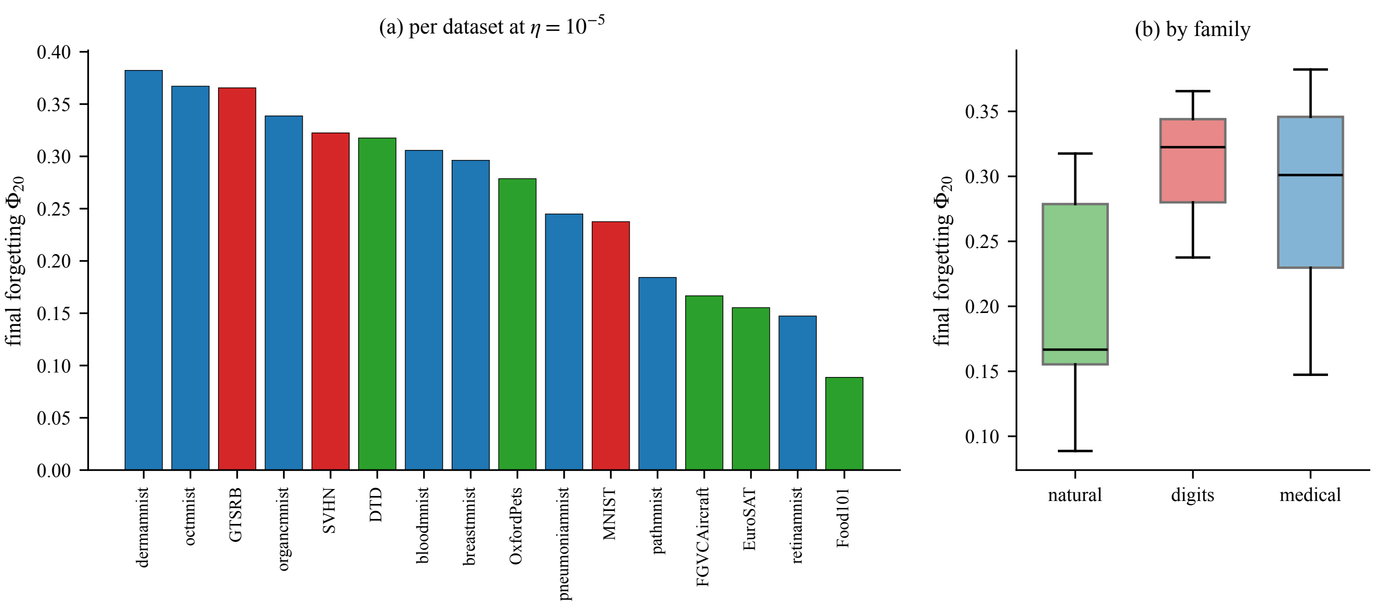

Fig. 2. (a) Final forgetting per dataset at $\eta = 10^{-5}$, coloured by family. (b) Distribution by family. The medical family is highly dispersed, so family membership alone is a weak predictor.

B. Grouped evaluation

The three learning-rate runs of one dataset share $\mathbf{s}_d$ exactly and are statistically dependent. Random row splitting would therefore leak dataset identity and inflate scores [9]. We use leave-one-dataset-out: for each $d^\star$ the forecaster is fitted on the 45 rows from the other 15 datasets and predicts the 3 held-out rows. Feature standardisation—and, for the LSTM, sequence normalisation—is fitted on the training datasets only.

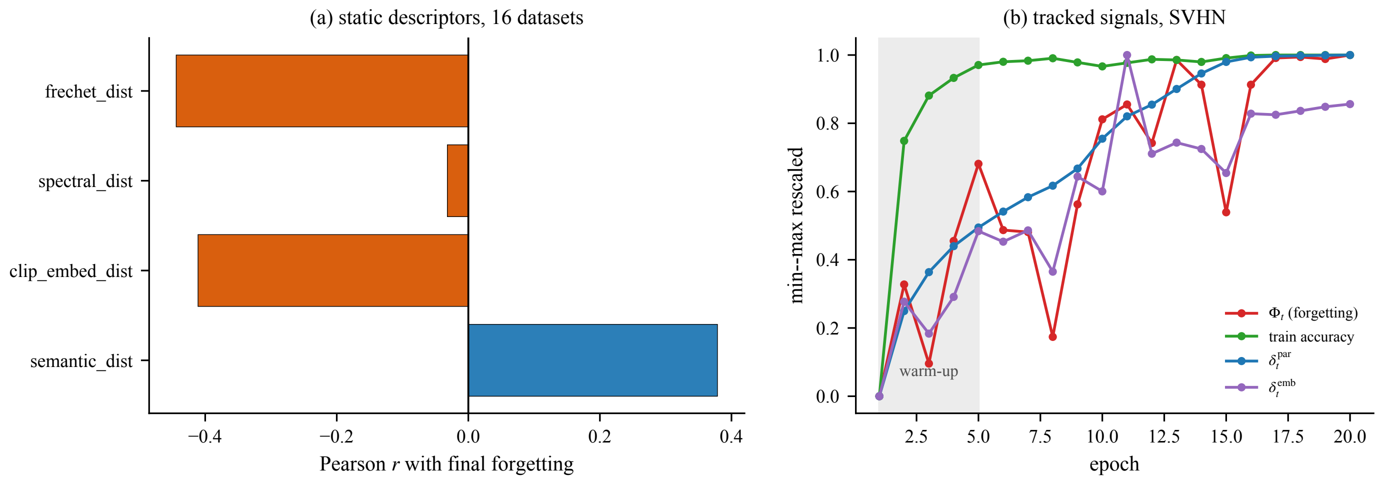

Fig. 3. (a) Pearson correlation of each static descriptor with final forgetting across the 16 datasets. All are modest and estimated from 16 points. (b) The four tracked signals for SVHN at $\eta = 10^{-5}$, min–max rescaled; the shaded region is the $K = 5$ warm-up.

C. Estimators and controls

Four tabular regressors are compared: ridge regression with generalised cross-validated penalty [10], a 300-tree random forest [11], gradient boosting with depth-2 stumps [12], and a $(32, 16)$ MLP. A single-layer LSTM [13] over the raw warm-up sequence, concatenated with the static block, is evaluated separately because it consumes the warm-up in ordered form. All are implemented in scikit-learn [14] except the LSTM.

We additionally define two model-free controls, which are the methodological core of this paper:

$$\text{Persistence:} \quad \hat{\Phi}_T = \Phi_K, \tag{7}$$

$$\text{Slope extrapolation:} \quad \hat{\Phi}_T = \Phi_K + \hat{\beta}_K \, (T - K). \tag{8}$$

Neither is fitted to anything. Any learned forecaster that consumes Φ_K must beat (7) to have demonstrated that it learned something beyond the fact that forgetting curves keep going.

Scores are mean absolute error, Spearman rank correlation, and $R^2 = 1 - \sum(\hat{\Phi} - \Phi)^2 / \sum(\Phi - \bar{\Phi})^2$, all computed over the pooled out-of-fold predictions.

V. Results

A. The warm-up ablation

Table II and Figure 4(a) give the central comparison. With static features alone, ridge regression reaches $R^2 = 0.426$ and the tree ensembles 0.29–0.39; this is genuine but weak signal, consistent with the modest univariate correlations of Figure 3(a). Adding the five-epoch warm-up moves all three well-behaved models to $R^2 \approx 0.91$–0.92 and reduces MAE from roughly 0.070 to 0.025, a 2.8× error reduction. Figure 4(b) shows the predicted-versus-actual scatter for the best model: held-out datasets from all three families lie close to the identity line, with no systematic family bias.

B. The persistence control changes the interpretation

Read without a control, Table II tells a strong story: watching five of twenty epochs more than doubles R^2. The persistence control reframes it. The rule "final forgetting equals forgetting at epoch 5" attains $R^2 = 0.848$, MAE 0.034, and—strikingly—the highest rank correlation in the entire table, $\rho = 0.945$. The best learned forecaster improves R^2 by 0.069 and MAE by 0.009, and is actually worse at ranking which dataset will forget most.

This is not a negative result about the study; it is a precise localisation of what the learned model contributes. Persistence systematically underestimates, because Φ_t is still rising at $t = 5$. The learned models supply the missing magnitude correction, which is exactly the kind of calibration a regression can learn from 45 neighbouring runs. What they do not supply is new information about the ordering of datasets, which the warm-up already encodes.

Slope extrapolation, Eq. (8), fails badly ($R^2 = -2.281$). Because the forgetting curves are concave, linearly continuing the early slope massively overestimates the endpoint. The failure is instructive: it shows that the warm-up's predictive content is in its level, not in its derivative, and that a plausible hand-crafted rule can be far worse than either the naive rule or the learned one.

C. How early is early enough?

Figure 5 sweeps the warm-up length K and re-runs the entire evaluation at each value, including both controls. Three regimes are visible.

For $K \leq 5$ the learned model leads clearly: at $K = 4$ it attains $R^2 = 0.848$ against persistence's 0.795, and at $K = 5$, 0.917 against 0.848. This is the operationally useful window, in which 75–80% of the run remains and aborting saves real compute. For $6 \leq K \leq 10$ the two curves interleave within ± 0.03 and neither dominates. For $K \geq 11$ both exceed $R^2 = 0.96$ and the forecasting

TABLE II

Leave-one-dataset-out forecasting of final forgetting Φ_{20}, $K = 5$. Rows above the rule are fitted models; rows below are model-free controls that consume no training data at all. Best fitted R^2 in bold.

Inputs	Estimator	MAE	Spearman ρ	R^2
static only	Ridge	0.070	0.683	0.426
	Random forest	0.070	0.670	0.394
	Grad. boosting	0.075	0.614	0.294
	MLP	0.122	0.346	−1.133
static + warm-up	Ridge	0.025	0.920	0.911
	Random forest	0.025	0.928	0.912
	Grad. boosting	0.025	0.924	0.917
	LSTM	0.038	0.878	0.812
	MLP	0.113	0.481	−0.660
control	Persistence, Eq. (7)	0.034	0.945	0.848
	Slope extrap., Eq. (8)	0.133	0.634	−2.281

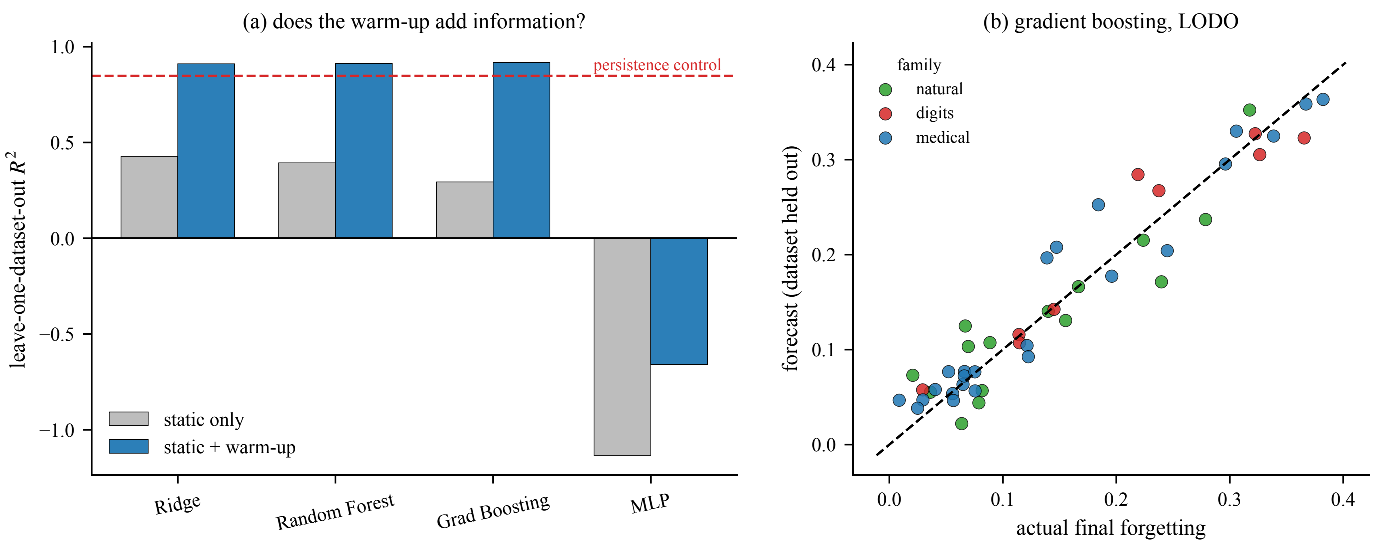

Fig. 4. (a) Leave-one-dataset-out R^2 with and without the warm-up block; the dashed red line is the persistence control of Eq. (7), which no static-only model approaches and which the warm-up models beat by only ≈ 0.07. (b) Predicted versus actual final forgetting for gradient boosting; each point is one held-out run.

problem has essentially dissolved: by then the curve has nearly saturated, so predicting the endpoint is close to reading it.

The practical recommendation follows directly. If the goal is an early-abort signal, use $K \approx 5$ and use a fitted model, because that is where fitting pays. If K can be larger, do not bother fitting anything.

D. Small samples punish flexible models

The MLP is worse than predicting the mean in both feature settings ($R^2 = -1.13$ and -0.66), and the LSTM, despite consuming strictly more information than the tabular models (the ordered sequence rather than its endpoint), scores 0.812 against gradient boosting's 0.917. With 45 training rows drawn from 15 independent groups, the effective sample size is far below what either architecture needs. This is a clean instance of a general rule: capacity is only an asset when the group count, not the row count, supports it.

VI. Discussion

A. What is supported

1) Final forgetting varies by more than a factor of four across datasets and learning rates in this campaign (Φ_{20} from 0.089 to 0.382 at $\eta = 10^{-5}$).
2) Early training dynamics carry substantially more forecast signal than the four static descriptors: R^2 rises from 0.43 to 0.92 under leave-one-dataset-out evaluation.
3) Most of that signal is persistence. A zero-parameter rule reaches $R^2 = 0.85$ and the best rank correlation observed. The learned contribution is a magnitude correction worth ≈ 0.07 R^2, concentrated at $K \leq 5$.
4) Simple regularised estimators dominate an MLP and an LSTM at this sample size.

B. Threats to validity

The independent unit is a dataset, so the effective sample size for a generalisation claim is 16, not 48; second and

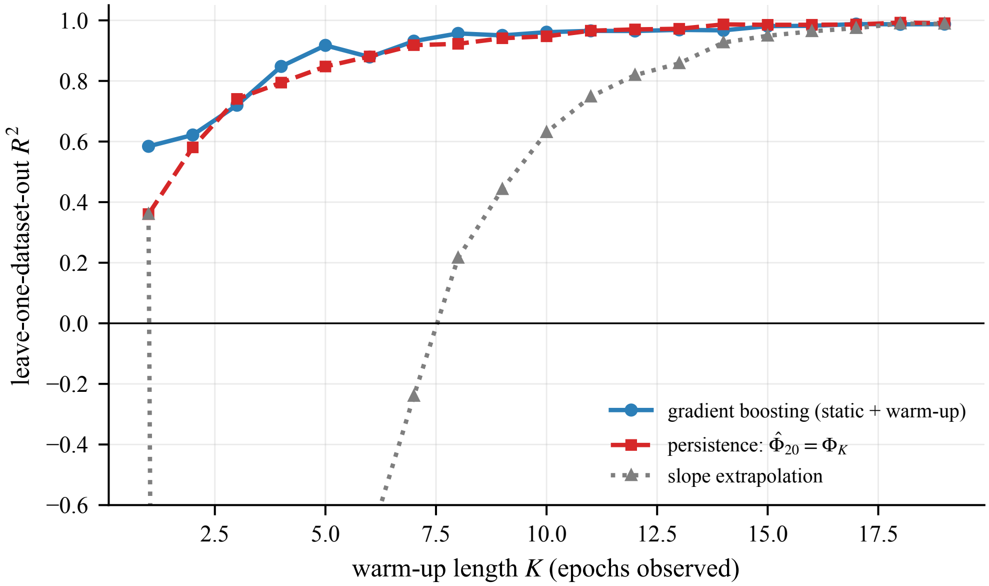

Fig. 5. Forecast quality against warm-up length K. The learned forecaster's advantage over persistence is confined to $K \leq 5$; from $K \approx 6$ onward the two curves are interchangeable. Slope extrapolation is catastrophic for small K and only becomes competitive once there is almost nothing left to forecast.

third decimal places in Table II are not stable evidence. The campaign uses one architecture, one optimiser, one prompt template, and one run per configuration, with no seed replication, so run-to-run variance is unmeasured and is confounded with between-dataset variance. The three general benchmarks in $\mathcal{B}$ are all small natural-image classification sets, so Φ measures loss of one particular kind of general competence rather than of general competence as such. Finally, the static descriptors and the warm-up features were designed together with the campaign, so their relative merit is not an unbiased comparison against the literature's static predictors.

C. Recommendations for future work

The decisive next experiment is not a better forecaster but a broader campaign: multiple seeds per configuration, at least two architectures, and at least two fine-tuning recipes (full fine-tuning versus parameter-efficient adaptation), evaluated against the persistence control of Eq. (7) at every warm-up length. We would also expect a genuinely stronger forecaster to use the drift signals (4)–(5) to anticipate the saturation point rather than to regress on the current level; that is the capability persistence provably lacks and which no model here demonstrably achieved.

VII. Conclusion

Watching the first five epochs of a CLIP fine-tuning run forecasts its final forgetting with $R^2 = 0.92$ on datasets the forecaster has never seen, more than doubling what pre-training descriptors achieve alone. But a rule that simply asserts the forgetting will not change reaches $R^2 = 0.85$ and ranks datasets better. The honest headline is therefore narrower and more useful than the raw ablation suggests: early dynamics forecast forgetting mainly because forgetting curves persist, and the value a fitted model adds is a calibrated magnitude correction in the short-warm-up regime where an early abort is still worth making.

Reproducibility

The campaign generator (trajectory_dynamics.py), the evaluation module (forecasting.py), the cached per-epoch logs, and the figure and table scripts accompanying this manuscript regenerate every number reported here in seconds on a CPU. Rebuilding the campaign from scratch requires a few GPU-hours via

```
python trajectory_dynamics.py --epochs 20 --train-n 4000 \
    --eval-n 1500 --lrs 1e-6 5e-6 1e-5
```

References

[1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal et al., "Learning transferable visual models from natural language supervision," in Proceedings of the 38th International Conference on Machine Learning (ICML), 2021, pp. 8748–8763.

[2] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," in Psychology of Learning and Motivation, vol. 24, 1989, pp. 109–165.

[3] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu et al., "Overcoming catastrophic forgetting in neural networks," Proceedings of the National Academy of Sciences, vol. 114, no. 13, pp. 3521–3526, 2017.

[4] M. Wortsman, G. Ilharco, J. W. Kim, M. Li, S. Kornblith, R. Roelofs et al., "Robust fine-tuning of zero-shot models," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 7959–7971.

[5] G. Ilharco, M. Wortsman, S. Y. Gadre, S. Song, H. Hajishirzi, S. Kornblith, A. Farhadi, and L. Schmidt, “Patching open-vocabulary models by interpolating weights,” Advances in Neural Information Processing Systems (NeurIPS), vol. 35, pp. 29 262–29 277, 2022.

[6] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in International Conference on Learning Representations (ICLR), 2019.

[7] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, “MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification,” Scientific Data, vol. 10, no. 1, p. 41, 2023.

[8] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, “GANs trained by a two time-scale update rule converge to a local Nash equilibrium,” in Advances in Neural Information Processing Systems (NeurIPS), 2017, pp. 6626–6637.

[9] G. C. Cawley and N. L. C. Talbot, “On over-fitting in model selection and subsequent selection bias in performance evaluation,” Journal of Machine Learning Research, vol. 11, pp. 2079–2107, 2010.

[10] A. E. Hoerl and R. W. Kennard, “Ridge regression: Biased estimation for nonorthogonal problems,” Technometrics, vol. 12, no. 1, pp. 55–67, 1970.

[11] L. Breiman, “Random forests,” Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[12] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” The Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001.

[13] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[14] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel et al., “Scikit-learn: Machine learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.